Genetic drift in mitochondrial segregation

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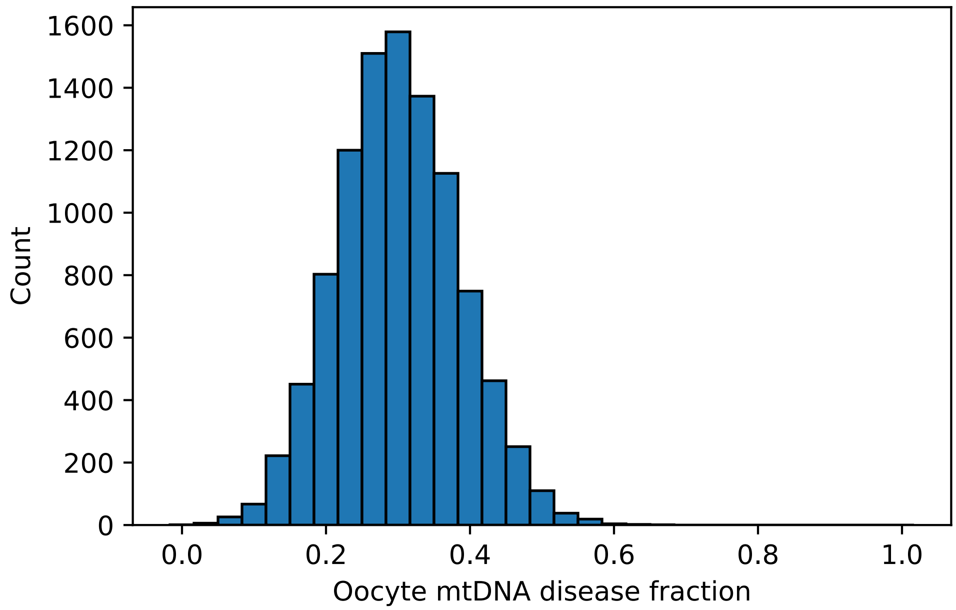
```python
try:
    import au_molecular_genetics
    print("Already installed")
except ImportError:
    %pip install -q "au_molecular_genetics @ git+https://github.com/au-mbg/molecular_
genetics.git"
```

```python
import numpy as np
import matplotlib.pyplot as plt
```

1 Exploring variation from mother to oocytes.

Even though the mother has a fixed mtDNA composition, random sampling during oocyte formation leads to substantial variation between oocytes, the simulation below explores this.

```python
def sample_oocyte(p, N, n_samples=1):
    oocyte_sample = np.random.binomial(n=N, p=p, size=n_samples)
    p_oocyte = oocyte_sample / N
    return oocyte_sample, p_oocyte


## Parameters
N = 30 # Effective bottleneck size
mother_disease_fraction = 0.3
n_samples = 10000

## Sampling
sample, p_ooc = sample_oocyte(mother_disease_fraction, N, n_samples)

## Plot
fig, ax = plt.subplots()
bins = np.linspace(-0.5/N, 1 + 0.5/N, N + 2)
ax.hist(p_ooc, bins=bins, edgecolor='black')
ax.set_xlabel('Oocyte mtDNA disease fraction')
ax.set_ylabel('Count')
plt.show()
```



Exercise 1

Run the simulation with `N=4`, `N=30`, `N=100` keeping `mother_disease_fraction=0.3` and `n_samples=10000`. What is the effect of increasing `N`? For which `N` do oocytes most often have mtDNA fractions close to the mother's value?

Exercise 2

Set `N=30` and vary `mother_disease_fraction` through `0.1`, `0.3` and `0.6`. How does the center of the distribution change? Does the width of the distribution change significantly?

Exercise 3

Two mothers have the same average mtDNA disease fraction. One produces children with very different outcomes, the other does not. Which model parameter explains this difference, and why?

2 Oocyte disease model

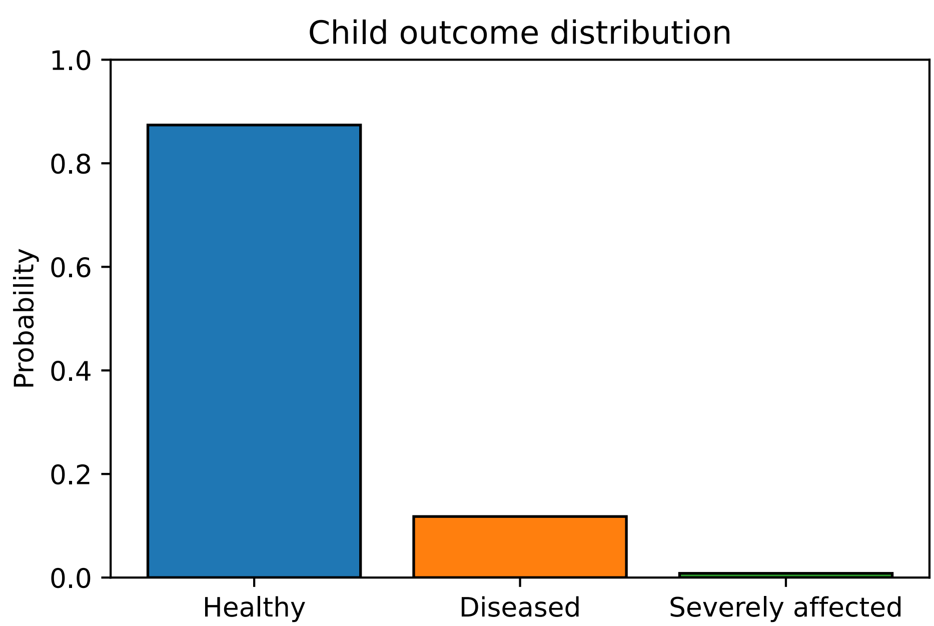
A simple model of the risk of disease development in a child based on the disease allele frequency in the oocyte is a threshold model. That is

- **Case 1:** Low disease allele frequency ($f < t_1$) $\rightarrow$ Healthy child
- **Case 2:** Intermediate disease allele frequency ($t_1 \leq f < t_2$) $\rightarrow$ Diseased child
- **Case 3:** High allele frequency in the oocyte ($f \geq t_2$) $\rightarrow$ Severely affected child

The cell below implements such a model, where you can control the parameters of oocyte sampling and the threshold model.

```
def threshold_model(p, t1, t2):
    state = np.zeros_like(p)        # Case 1
    state[(p >= t1) & (p < t2)] = 1 # Case 2
    state[p >= t2] = 2              # Case 3
    return state
```

```python
## Parameters
N = 30
mother_disease_fraction = 0.2
n_samples = 10000

t1 = 0.3 # Disease threshold
t2 = 0.4 # Severe threshold

## Sampling
sample, p_ooc = sample_oocyte(mother_disease_fraction, N, n_samples)

child_state = threshold_model(p_ooc, t1, t2)
states = {0: 'Healthy', 1: 'Diseased', 2:'Severely affected'}

fig, ax = plt.subplots()

for state, desc in states.items():
    ax.bar(state, np.sum(child_state == state)/n_samples, width=0.8, edgecolor='black')

ax.set_xticks(list(states.keys()))
ax.set_xticklabels(list(states.values()))
ax.set_ylabel('Probability')
ax.set_title('Child outcome distribution')
ax.set_ylim([0, 1])
plt.show()
```



Exercise 4

For `N=30` and `mother_disease_fraction = 0.2`, `t1 = 0.3` and `t2=0.4`, the mother's mtDNA disease fraction is below the disease threshold. Why do diseased or severely affected children still appear in the simulation?

Exercise 5

Keeping `mother_disease_fraction = 0.2`, `t1 = 0.3`, `t2 = 0.4`, compare the outcome probabilities for `N = 4`, `30`, and `100`. Which `N` gives the highest probability of severe outcomes, and why?